

SkateGuard Project Progress Report

Weeks 3 and 4

1. Report Scope

Weeks 3 and 4 moved the project from hardware bring up into the embedded control layer. The main engineering work was the motion state machine, kickflip and wobble detection, GPS validity handling, ESP NOW telemetry and the wrist alert path.

2. Embedded Motion Detection

Kickflip detection was implemented as a four stage state machine. The controller waits in IDLE, enters ROTATING when rotational motion crosses the trigger condition, accumulates rotation during the active window, and then enters LAND_CONFIRM to determine whether the deck settles inside the permitted range. COOLDOWN prevents immediate retriggering after a counted event.

2.1 Motion Detection Parameters

Parameter	Value	Purpose / implementation
Motion trigger	150 deg/s	Rotational rate required to start active motion
Active motion window	2000 ms	Normal rotation accumulation window
Absolute motion timeout	4000 ms	Hard limit for one motion event
Land confirmation window	1000 ms	Time allowed to confirm deck settling
Land confirmation limit	83 deg	Maximum settling swing for a confirmed land
Bail threshold	Outside 83 deg settling range	Classifies the motion as bail
Cooldown	500 ms	Quiet period after a counted event
Wobble oscillations required	3	Oscillation count required to confirm wobble
Wobble speed gate	None	Speed is not used as the trigger

3. Kickflip State Machine

- IDLE: the system waits for significant rotational motion.
- ROTATING: gyro Y is accumulated over time and the motion window is tracked.
- LAND_CONFIRM: after sufficient rotation, the system watches the deck swing during the landing interval.
- COOLDOWN: the detector remains quiet briefly before accepting another event.

A flip counts as a land only if the deck settles within 83 degrees

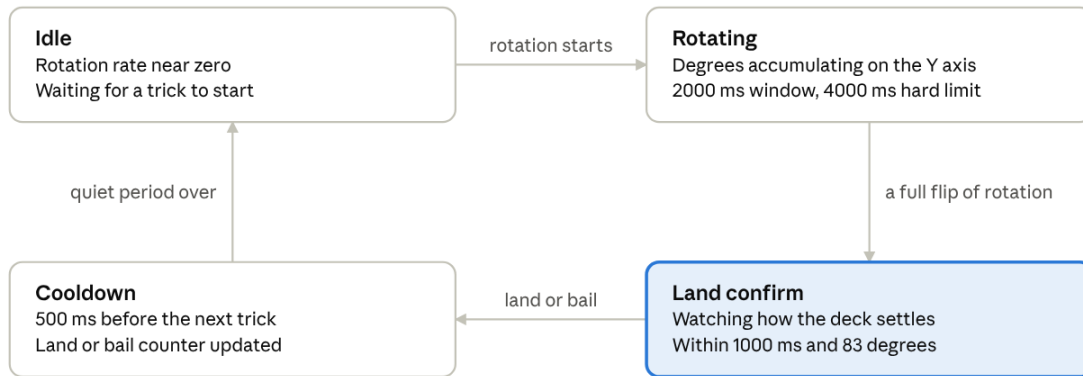


Figure 1. Kickflip detection state machine: Idle, Rotating, Land confirm and Cooldown.

The current firmware uses GYRO_Y as both WOBBLE_AXIS and KICKFLIP_AXIS. This is explicitly a mounting assumption. The stage log also records gyro X and gyro Z so that a real kickflip can be used to confirm the dominant axis.



Debug	
Stage	IDLE
Gyro Y	-0.9 deg/s
Accum Rotation	0.0 deg
Land Confirm Swing	0.0 deg
Active Window	0 / 2000 ms
Absolute Window	0 / 4000 ms
Cooldown Window	0 / 500 ms
Oscillations	0 / 3
Loop Max / Avg	12 / 3.8 ms
Land Count	1
Bail Count	0
Battery	86% (4.02 V)
GPS Satellites	9
GPS HDOP	0.9
GPS Fix Age	0.4 s
GPS Data In	9426 chars
GPS Checksum Fails	0
WiFi	ch 6 -54 dBm
ESP-NOW	wearable linked
Free Heap	175 KB
Uptime	9s

Download stage_log.csv Clear log

Figure 2. Debug page on the running board showing the motion stage, gyro Y, accumulated rotation, land confirmation swing, oscillation count, land/bail counters and GPS status.

4. GPS Processing

GPS data is not treated as valid merely because TinyGPS++ has previously received a location. The firmware requires a current valid position, a position age below the allowed limit and at least four satellites. Speeds below 2 km/h are reported as zero to prevent stationary GPS drift from becoming ride motion. Implausible position jumps are rejected using a speed based plausibility check.

4.1 GPS Validation Parameters

Parameter	Value	Purpose / implementation
GPS interface	UART1, 9600 baud	NEO 8M data input
Minimum satellites	4	Required for trusted position
Maximum position age	< 3 s	Prevents stale GPS data
Stationary speed cutoff	2 km/h	Suppresses small GPS drift
Implausible movement check	20 m/s reference	Rejects multipath or bad position jumps
GPS update behavior	Position update based	Distance is accumulated only from accepted positions

5. ESP NOW Communication

The board broadcasts telemetry using ESP NOW while maintaining its phone hotspot. The wrist unit does not assume the hotspot channel at startup. It scans channels for 300 ms intervals until a board packet is received, reads the board channel from telemetry and retunes accordingly. The telemetry packet includes tilt angle, speed, distance, board IP, GPS fix state, wobble state, kickflip state and board channel.

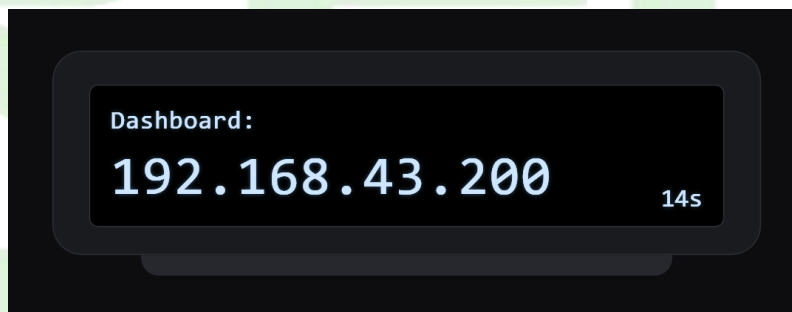


Figure 3. Wrist unit OLED showing the board dashboard address received over ESP NOW.

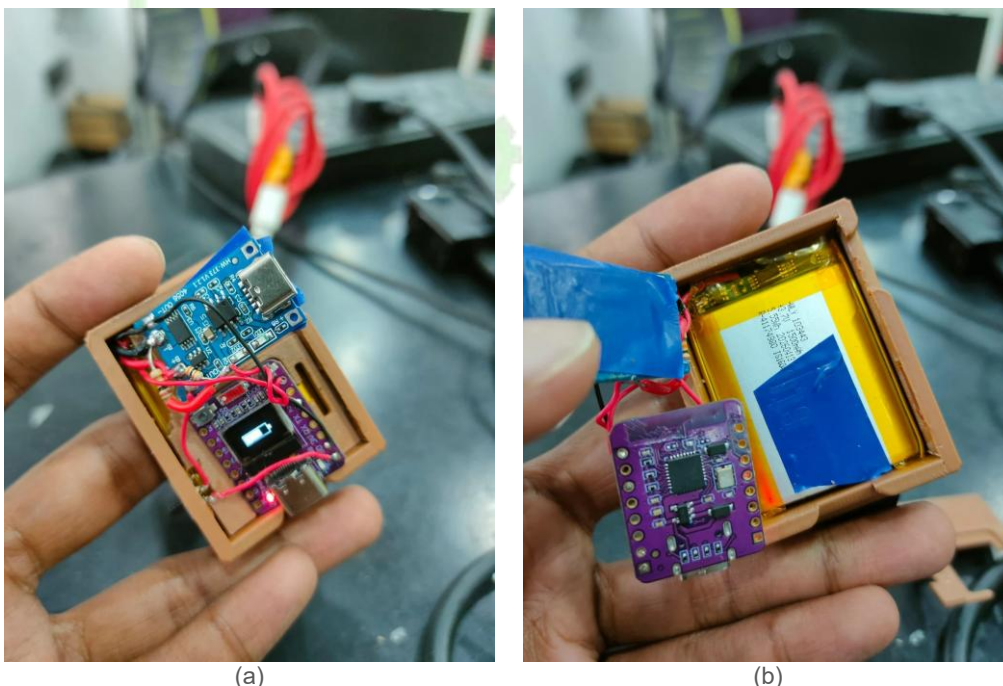


Figure 4. Wrist alert unit hardware: (a) controller, OLED and charging hardware, (b) battery installation.

5.1 Wireless and Haptic Parameters

Parameter	Value	Purpose / implementation
ESP NOW peer	Broadcast FF:FF:FF:FF:FF:FF	Board telemetry broadcast
Peer channel	0	Follows the station interface channel
Channel hunting	300 ms per channel	Wrist discovery procedure
Telemetry interval	100 ms	Board to wrist update rate
Wobble screen latch	500 ms	Visible event indication
Wobble ongoing period	1500 ms	Wrist event handling duration
Wrist motor	GPIO3	Non blocking haptic pulse
Wrist LED	GPIO8	Visual event indicator

6. Important Firmware Issue and Resolution

During wireless development, serial printing inside the ESP NOW receive callback could block the WiFi task long enough to make a healthy radio connection appear unreliable. The receive callback was changed to place the information into a queue, while the actual printing was performed outside the callback. This reduced interference with packet reception.

7. Bail Detection Change

The original approach relied on an accelerometer impact spike. This was rejected because both a clean landing and a bail can produce inconsistent impact signatures. The final approach evaluates how far the deck swings after the rotation. A settling response inside the 83 degree limit during the 1000 ms confirmation period is counted as a land; excessive or slow settling becomes a bail.

8. Detection Test Dataset

8.1 Kickflip Detection Test Data

Sample	Timestamp ms	Stage	Gyro y dps	Accum deg	Land confirm swing deg	Event	Kickflip count	Bail count
1	0	IDLE	5.780	0.000	0.000	0	0	0
2	100	IDLE	8.730	0.000	0.000	0	0	0
3	200	IDLE	-1.200	0.000	0.000	0	0	0
4	300	IDLE	4.690	0.000	0.000	0	0	0
5	400	IDLE	1.050	0.000	0.000	0	0	0
6	500	IDLE	1.540	0.000	0.000	0	0	0
7	600	IDLE	1.220	0.000	0.000	0	0	0
8	700	IDLE	0.410	0.000	0.000	0	0	0
9	800	IDLE	3.150	0.000	0.000	0	0	0
10	900	IDLE	0.510	0.000	0.000	0	0	0
11	1000	IDLE	3.810	0.000	0.000	0	0	0
12	1100	IDLE	2.850	0.000	0.000	0	0	0

Table shows the first 12 rows from the test dataset for traceability. The complete CSV is retained separately.

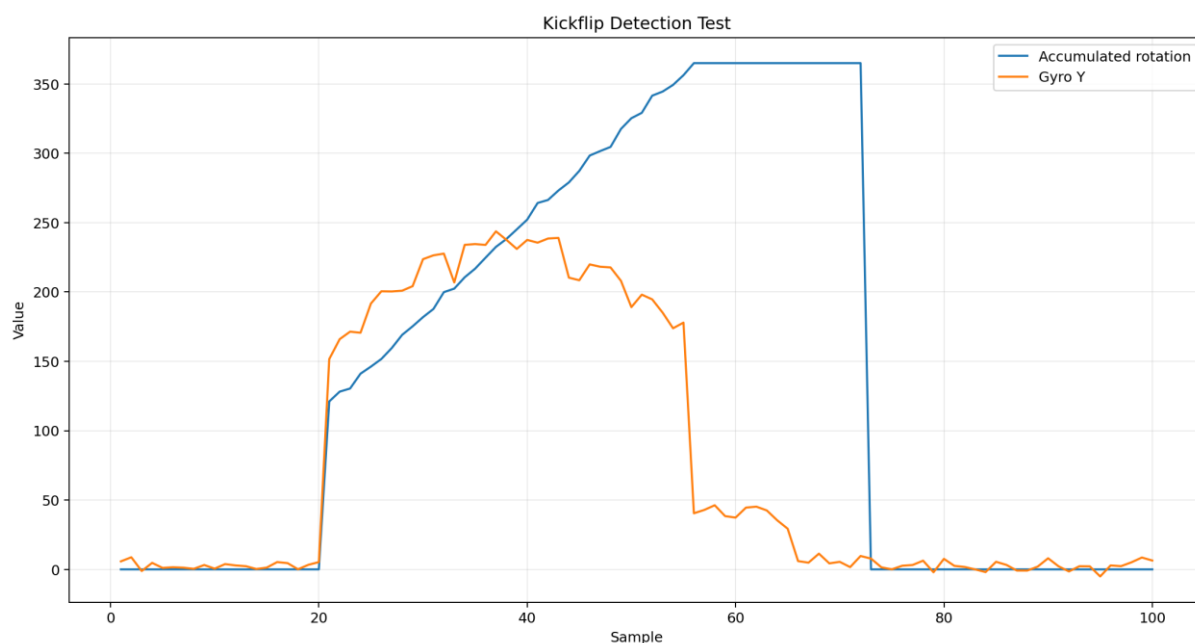


Figure 5. Rotational response and accumulated angle used to visualize the kickflip detection sequence.

9. Wobble Test Dataset

9.1 Wobble Detection Test Data

Sample	Timestamp ms	Gyro y dps	Speed kmh	Reversal streak	Oscillation count	Wobble active	Motion state
1	0	2.280	11.420	0	0	0	IDLE
2	100	6.510	12.260	0	0	0	IDLE
3	200	-1.520	12.240	0	0	0	IDLE
4	300	-0.810	12.000	0	0	0	IDLE
5	400	-2.330	12.660	0	0	0	IDLE
6	500	-4.060	13.790	0	0	0	IDLE
7	600	-2.600	13.420	0	0	0	IDLE
8	700	-4.900	9.770	0	0	0	IDLE
9	800	0.140	8.170	0	0	0	IDLE
10	900	-3.080	13.400	0	0	0	IDLE

Table shows the first 10 rows from the test dataset for traceability. The complete CSV is retained separately.

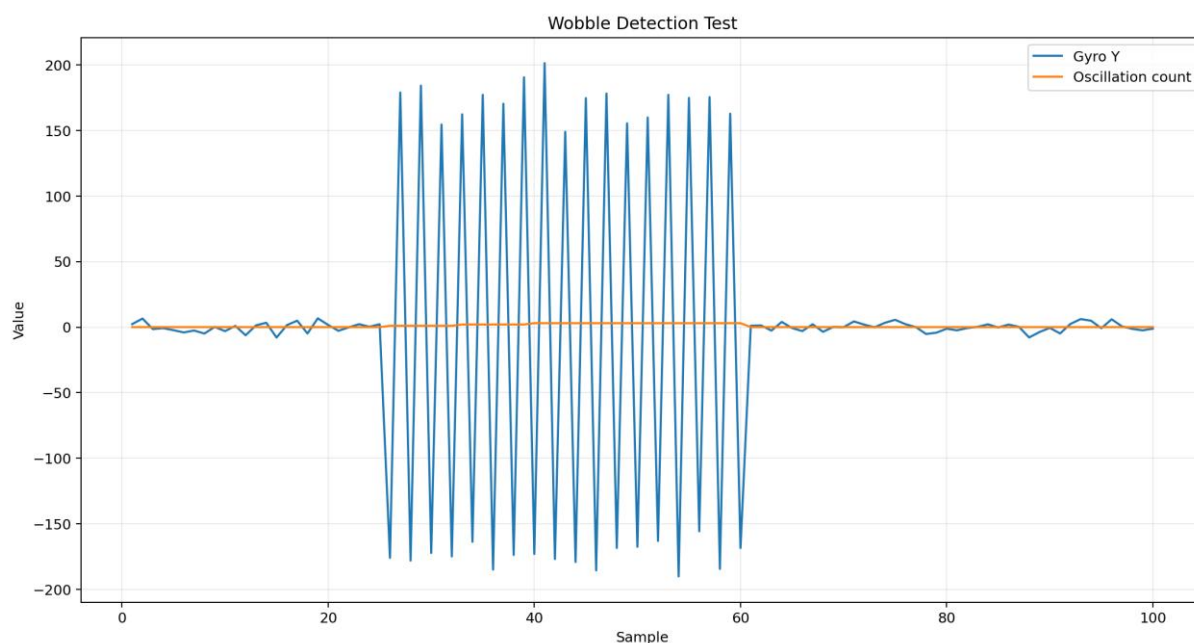


Figure 6. Wobble test showing gyro response and counted oscillations.

10. Week 4 Status

By the end of Week 4, the core embedded detection logic, board to wrist communication, haptic alert path and bail classification logic were implemented. Field confirmation of the assumed rotation axis and real riding speed wobble remained open.